Distracted Driver Detection on Unseen Drivers

Name: Lokaraj R **Register No.:** RA2311026050005 **Section:** AIML B
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GitHub: https://github.com/LokarajR/Distracted-Driver-Detection-PS48

1. Problem and Objective

Driver distraction is a major cause of road accidents. The task is to classify an in-car camera image into ten driver behaviours (c0 safe driving … c9 talking to passenger) so a vehicle could warn the driver. It must work on new drivers, so success is judged by macro F1 on test drivers held out from training (every behaviour counts equally), plus accuracy, per-class precision/recall, confusion matrix, top-2 accuracy and log loss (the Kaggle metric).

2. Dataset and Pre-processing

State Farm Distracted Driver Detection (Kaggle competition): 22,424 labelled 640 × 480 images of 26 drivers; classes are nearly balanced (1,911 to 2,489 images, ratio 1.30). Following the lab note for large datasets, a fixed balanced subset of up to 16 images per (driver, class) cell was drawn (seed 42): 4,136 images. Because each driver was filmed in one session, frames of the same driver are near-duplicates, so the split is grouped by driver: 17 drivers for training (2,720 images), 4 for validation (616) and 5 for testing (800; p002, p026, p041, p050, p075). No driver appears in two splits (asserted; saved to outputs/splits/). Images were resized to 224 × 224 and ImageNet-normalised. Training-only augmentation: random resized crop (scale 0.75–1), rotation ±10°, colour jitter, random erasing; no horizontal flip, as it would turn texting-right (c1) into texting-left (c3). The balanced subset needs no re-weighting.

3. Methodology

Four CNNs were compared: a 5-block CNN trained from scratch (2.36 M parameters) and three ImageNet-pretrained backbones, MobileNetV3-Large (4.21 M), EfficientNet-B0 (4.02 M) and ResNet-18 (11.18 M). Pretrained models got a new 10-class layer, trained for two epochs with the backbone frozen, then the whole network was fine-tuned at a lower learning rate. All models used cross-entropy with label smoothing 0.1, AdamW with a one-cycle schedule, and early stopping on validation macro F1 (best epoch restored). The model was selected on validation drivers only; the test drivers were used once. As a leakage control, the selected model was retrained identically on a random image-level split of the same size.

Table 1. Hyper-parameters.

Setting	Value	Setting	Value
Input size / batch size	224 × 224 / 32	Optimiser / schedule	AdamW, one-cycle LR
Learning rate (head / fine-tune)	1×10^{-3} / 3×10^{-4}	Learning rate (from scratch)	1×10^{-3}
Epochs (pretrained)	2 head + ≤ 10 fine-tune	Epochs (from scratch)	≤ 20
Early stopping	patience 4 on val. macro F1	Weight decay / label smoothing	1×10^{-4} / 0.1
Loss	cross-entropy	Seed / framework	42 / PyTorch 2.11, Colab GPU

4. Results

ResNet-18 had the best validation macro F1 (0.85) and was selected. On the 800 images of the five unseen test drivers it reached 89.75 % accuracy, macro F1 0.90, top-2 accuracy 96.00 % and log loss 0.48 (Table 2). Per-driver test accuracy ranged from 86.88 % (p002) to 93.12 % (p026). Eight classes reached F1 ≥ 0.86, but talking to passenger (c9) had recall 53.75 % (F1 0.65) and safe driving (c0) precision 63.54 % (Fig. 1a). The same network trained on a random split scored 98.75 %.

Table 2. Test results (800 images, 5 unseen drivers, 80 per class). Latency: Colab CPU, batch 1. Selected model shaded; last row is the leakage control.

Model	Params (M)	CPU latency (ms/img)	Train (min)	Val macro F1	Test acc. (%)	Test macro F1	Test log loss	Top-2 acc. (%)
Scratch CNN	2.36	89.81	7.94	0.76	73.75	0.72	0.82	86.88
MobileNetV3-Large	4.21	20.20	3.28	0.70	78.00	0.77	0.80	89.75
EfficientNet-B0	4.02	35.56	3.60	0.76	85.50	0.85	0.54	94.38
ResNet-18 (selected)	**11.18**	**59.08**	**4.54**	**0.85**	**89.75**	**0.90**	**0.48**	**96.00**
ResNet-18, random split	11.18	59.08	–	1.00	98.75	0.99	0.14	–

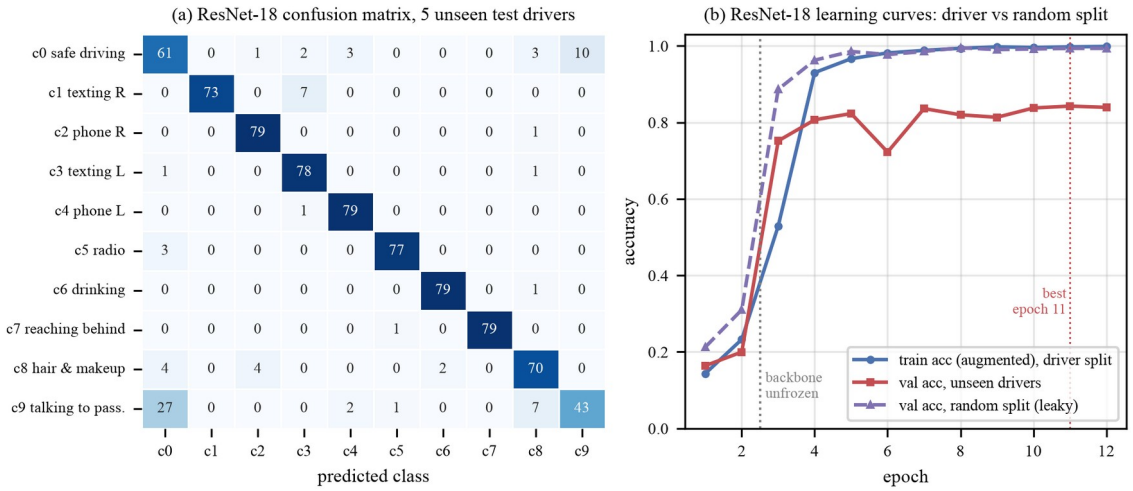


Fig. 1. (a) Confusion matrix of ResNet-18 on the test drivers (counts; colour = % of the true class). (b) Training and validation accuracy of ResNet-18 on the driver-grouped split, with validation accuracy of the same model on the random split.

5. Analysis and Discussion

The specific challenge, leakage through the driver, is clear: 50.63 % of random-split test images have a training image within RMS pixel distance 0.05, against 0.00 % for the driver split. The random split therefore reports 98.75 % accuracy against 89.75 % on new drivers, overstating performance by 9.00 points and cutting log loss from 0.48 to 0.14. In Fig. 1b the random-split validation curve follows the training curve (99.9 %), while unseen drivers plateau near 84 %: the gap is over-fitting to the 17 training drivers' appearance, which augmentation and early stopping reduce but do not remove. The most confused pair is talking to passenger → safe driving (27 of 80): both have both hands on the wheel, and only the head turn differs. Grad-CAM (Fig. 2, #2) shows the model looking at the arms and wheel, not at the face. Other errors (Fig. 2) share poses: a phone being lowered from the ear looks like texting (#1); a cup at the mouth with the elbow raised looks like touching hair (#3); a hand gesturing near the console looks like using the radio (#7); a hand on the cheek looks like holding a phone (#8). A single frame cannot show whether a hand is moving to or from the face. In a robustness test without retraining, macro F1 fell by 0.01 in low light (brightness × 0.5), 0.07 at low resolution (80 × 60) and 0.10 under blur ($\sigma = 2$).



Fig. 2. Misclassified test images (true → predicted, confidence) with Grad-CAM for the predicted class.

6. Real-world Relevance, Limitations and Ethics

The model could drive an in-car warning for phone use (F1 ≥ 0.93). It is not deployable yet: talking to a passenger is mostly missed; test results rest on only five drivers in daylight with one camera position; and a single frame cannot judge how long a driver was distracted. Missed distraction can cost lives and false alarms get ignored, so alerts should need several consecutive frames. In-car video is sensitive personal data: process it on the device, store nothing, and never use it against drivers without consent. The 26 drivers do not represent all ages, ethnicities or vehicles, so accuracy may be lower for some groups.

7. Conclusion and Future Work

With a driver-grouped split, a fine-tuned ResNet-18 recognises ten driver behaviours on five unseen drivers with 89.75 % accuracy (macro F1 0.90), while a random split would have reported a misleading 98.75 %. Next: add a short frame sequence or a head-pose crop to separate talking to the passenger from safe driving.

Dataset: State Farm Distracted Driver Detection, Kaggle, 2016, kaggle.com/c/state-farm-distracted-driver-detection. **Other:** Grad-CAM, Selvaraju et al., ICCV 2017. **AI declaration:** an AI assistant (Claude, Anthropic) helped write the notebook code and draft this report; every number comes from the executed notebook.